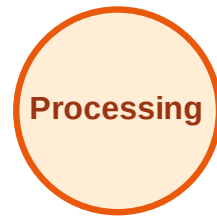


BFS Traversal

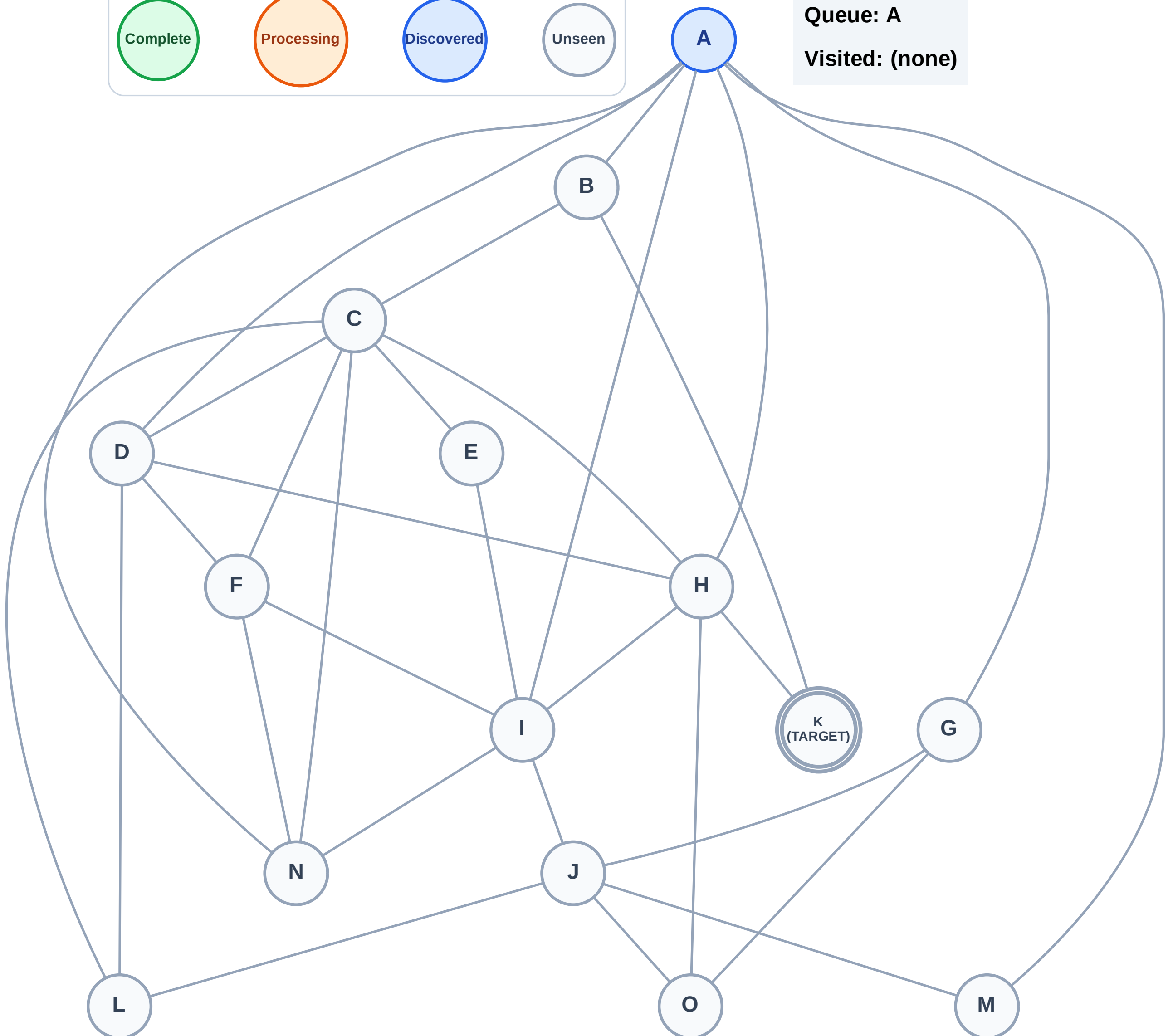
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

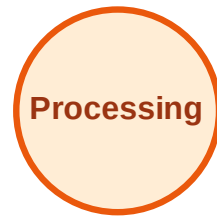
Visited: (none)



BFS Traversal

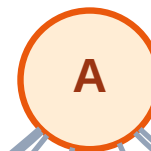
Step 2: Process node A

Legend



Queue: (empty)

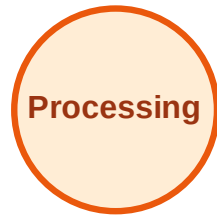
Visited: (none)



BFS Traversal

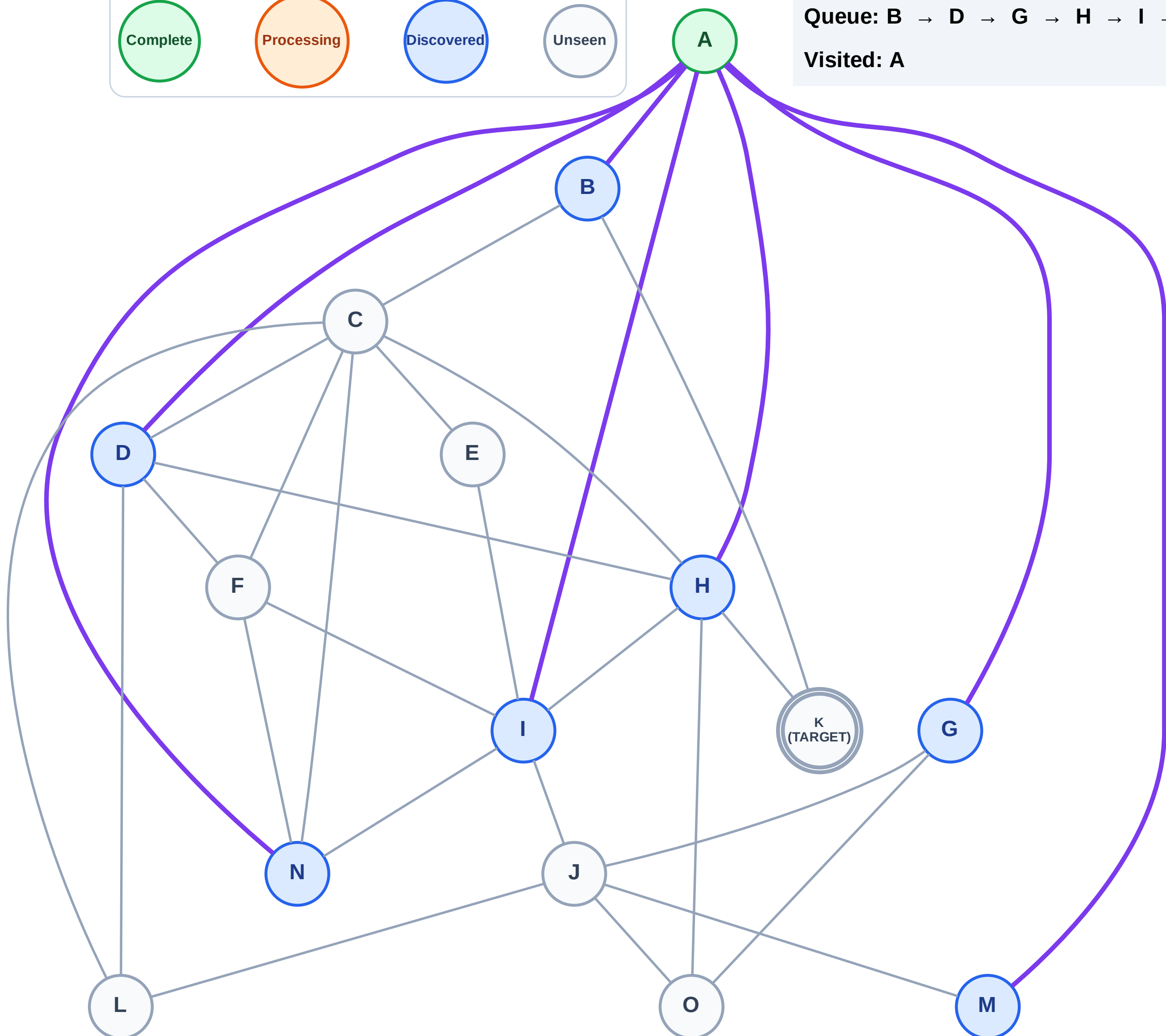
Step 3: Discover B, D, G, H, I, M, N from A

Legend



Queue: B → D → G → H → I → M → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

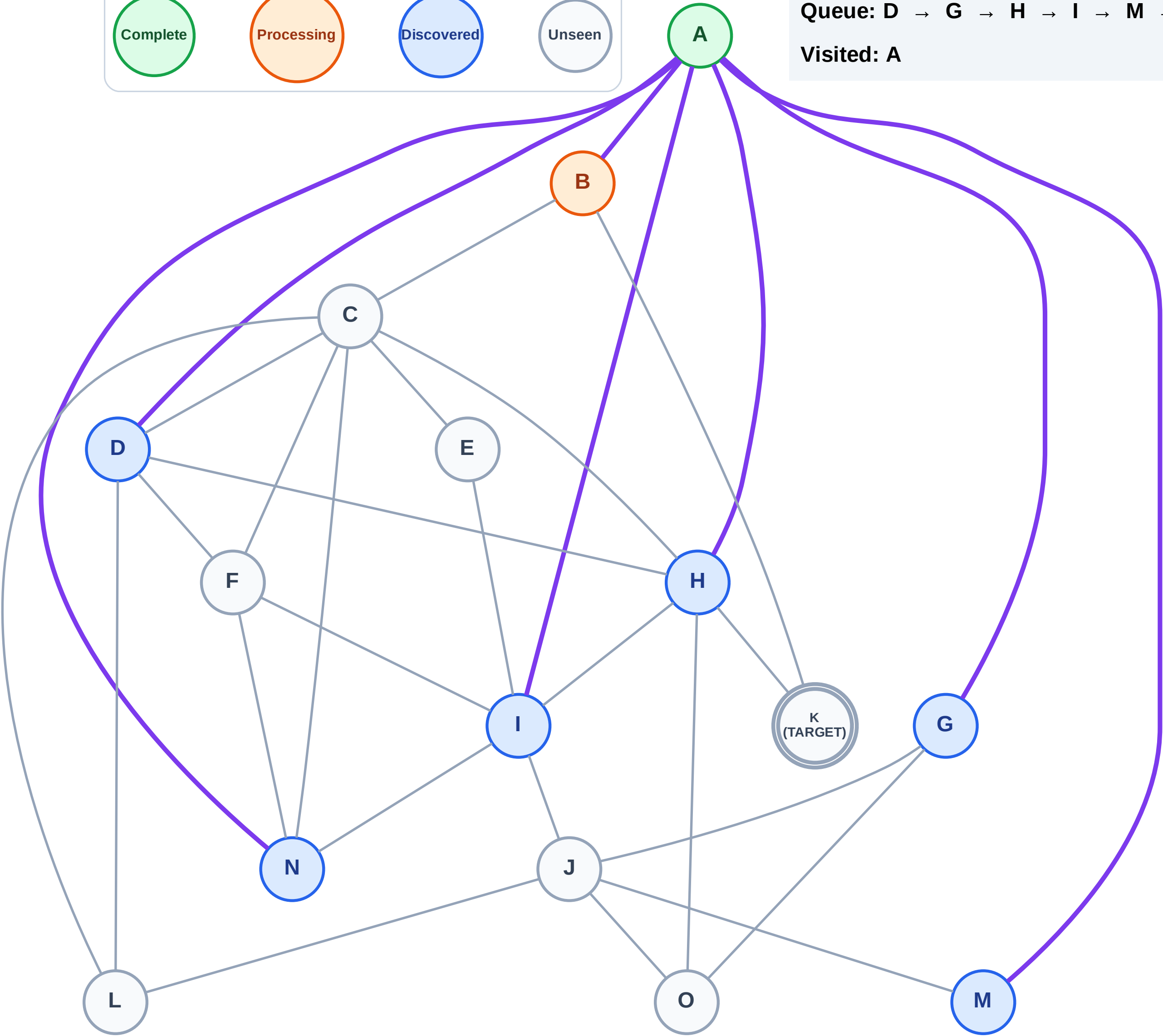
Processing

Discovered

Unseen

Queue: D → G → H → I → M → N

Visited: A



BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

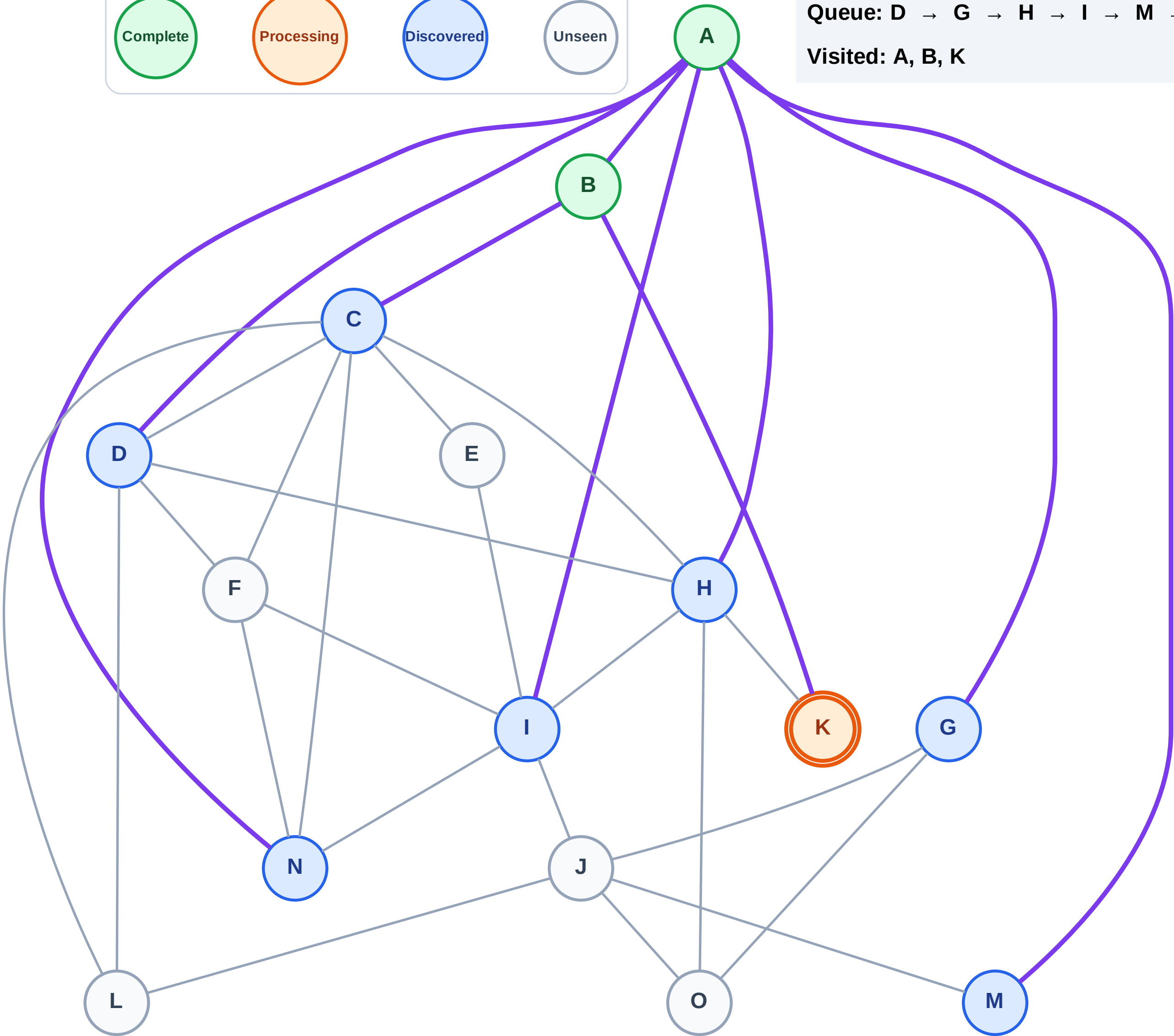
Processing

Discovered

Unseen

Queue: D → G → H → I → M → N → C

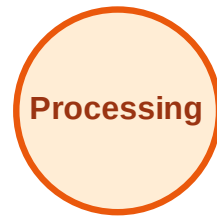
Visited: A, B, K



DFS Traversal

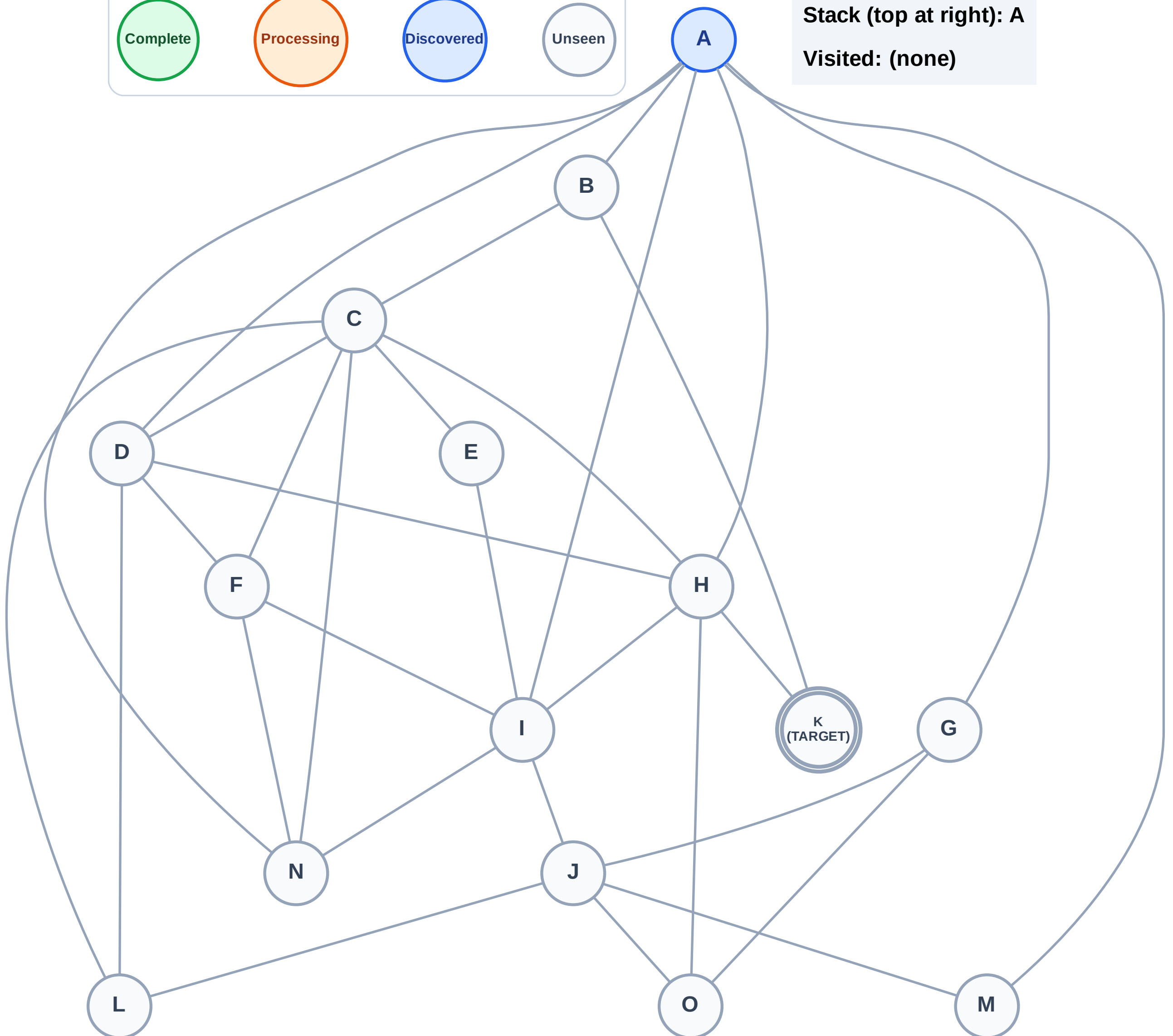
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

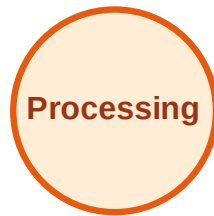
Visited: (none)



DFS Traversal

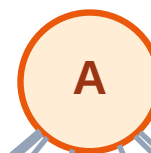
Step 2: Process node A

Legend



Stack (top at right): (empty)

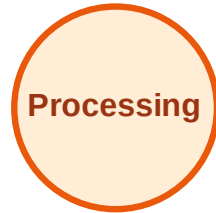
Visited: (none)



DFS Traversal

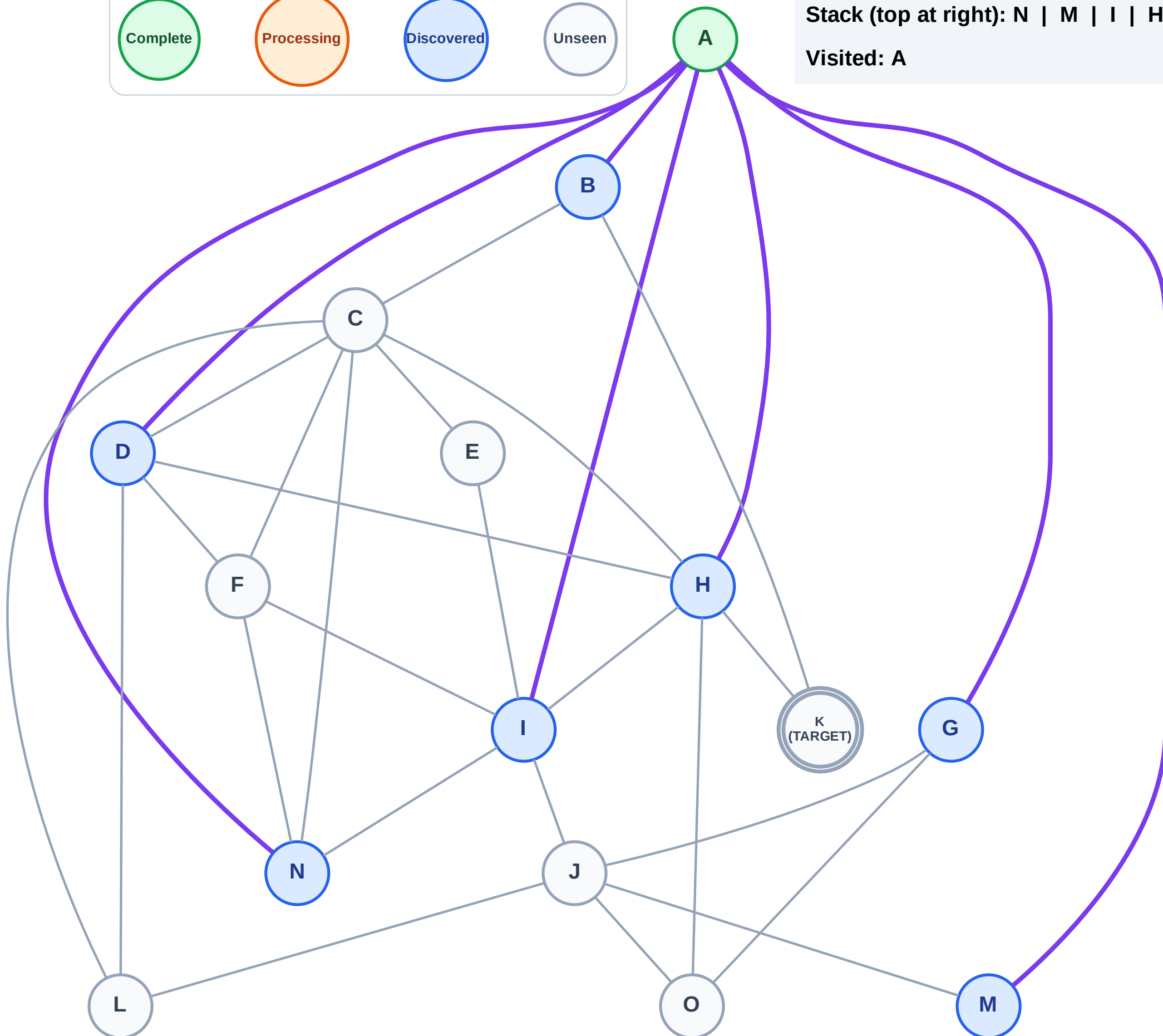
Step 3: Discover N, M, I, H, G, D, B from A

Legend



Stack (top at right): N | M | I | H | G | D | B

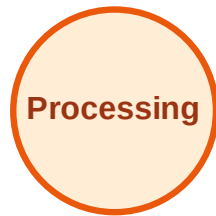
Visited: A



DFS Traversal

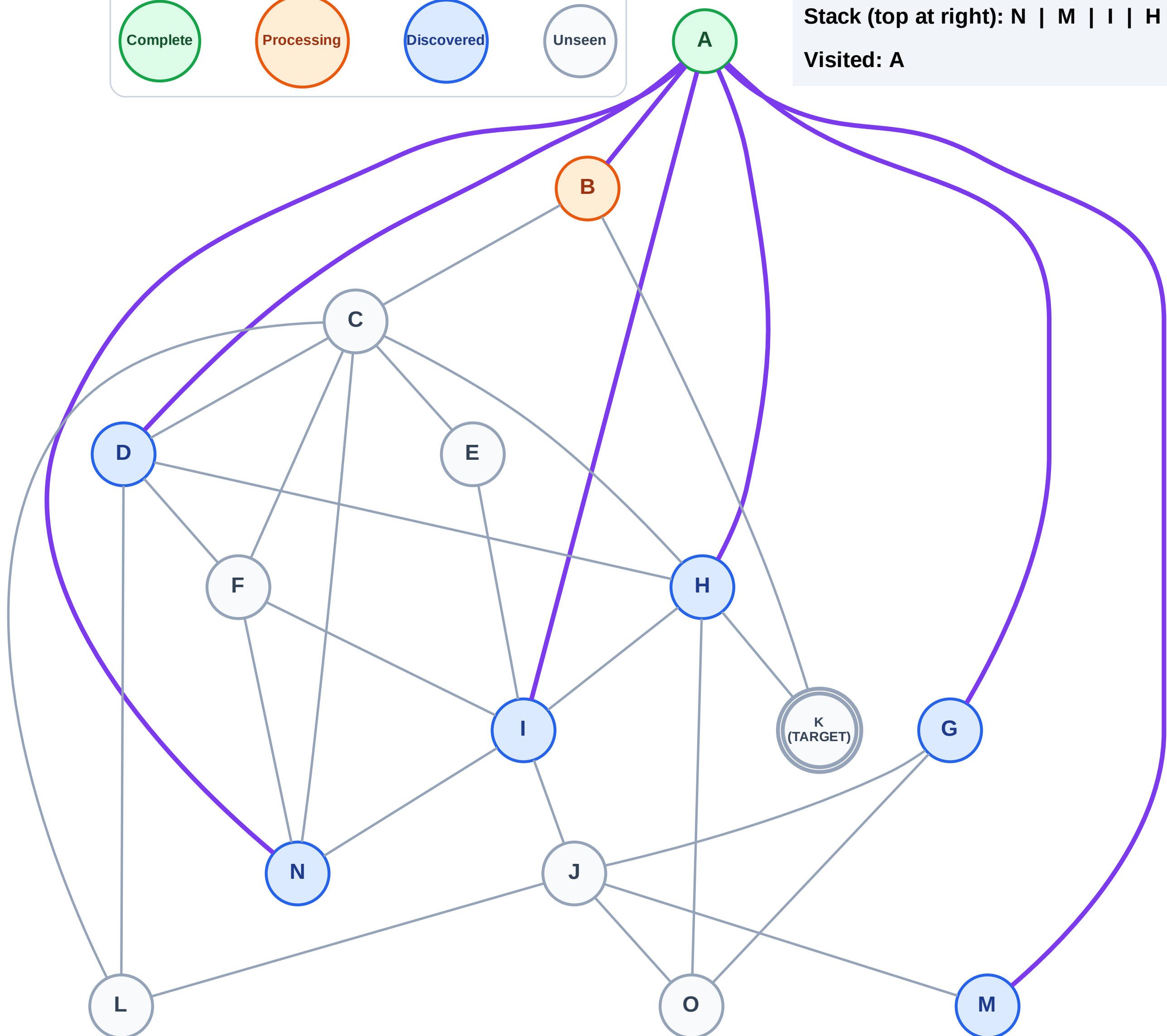
Step 4: Process node B

Legend



Stack (top at right): N | M | I | H | G | D

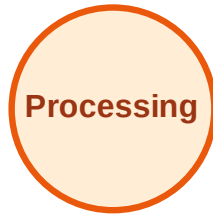
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): N | M | I | H | G | D

Visited: A, B, K

